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Probability Problems

1. total # of ways to ask 8 questions to 15 students

$- 15^8$

Select 8 students from 15 and arrange in order

$${}^{15}P_8 = \frac{15!}{\cancel{7!}} = 15 \cdot 14 \cdot \ldots \cdot 8 \cdot 7 = 259{,}459{,}200$$

probability no student will have to answer more than one question

$$\frac{259{,}459{,}200}{15^8} = 0.101237$$

2. total # of even integers that start with 2 odd digits

0 - 100 none

100 - 1000 $= 5 \cdot 4 \cdot 5 = 100$

1000 - 10,000 $= 5 \cdot 4 \cdot 7 \cdot 5 = 700$

10,000 - 99,999 $= 5 \cdot 4 \cdot 7 \cdot 6 \cdot 5 = 4200$

total $= 4200 + 700 + 100 = 5000$ numbers that meet criteria

probability of getting one of these #s.

$$- \frac{5000}{\text{total \#s}} = \frac{5,000}{100,000} = \boxed{0.05}$$

Generate 8 random #s

X = the # of integers that meet our criteria from 8

$$p(X=5) = {}^{8}C_{5} (0.05)^{5} (1-0.05)^{3}$$

$$= \boxed{1.5004 \times 10^{-5}}$$

3. probability of rolling ≥ 4 on dice: $\frac{3}{6}$

$\cdot P(A) = P(\geq 2 \text{ dice roll } \geq 4)$

$= P(X=2) + P(X=3) = 3C_2 \cdot (3/6)^2 \cdot (3/6) + 3C_3 \cdot (3/6)^3 \cdot (3/6)$

$= 1/2$

$\cdot P(B) = P(\text{All 3 dice roll the same \#})$

$= P(6 \text{ ways b/c there are } 6 \text{ \#s}) = 6/6^3 = 1/36$

$\cdot P(A \cap B) = P(\text{all 3 same } \geq 4)$

$= P(\text{all 3 \#s} = 4) + P(\text{all } 3 = 5) + P(\text{all } 3 = 6)$

$= 1/6^3 + 1/6^3 + 1/6^3 = 3/216 = 1/72$

$P(A) \cdot P(B) = (1/2) \cdot (1/36) = 1/72$ which is equal to $P(A \cap B)$, $\therefore$ events A & B are independent.

4. Poker Question

Probability of flush on a 5-card hand

$$- p = \frac{4 \cdot \binom{13}{5}}{\binom{52}{5}} = 0.001980792$$

let x = number of hands to play until one flush;

Each deal is from a new deck, it's sampling with replacement, so

x is Geometric$(p = 0.001980792$

$$E[X] = 1/p = \boxed{504.8486}$$

expected number of games to get a flush

5. Basketball Question

$P(\text{Win} \mid \text{super star plays}) = 0.7$

$P(\text{win} \mid \text{no superstar}) = 0.5$

$P(\text{superstar plays}) = 0.75$ for next 5 games

$\therefore P(\text{win } 4/5 \mid \text{superstar plays}) = \binom{5}{4} \cdot 0.7^4 \cdot 0.3$

$= 0.36015$

$P(\text{win } 4/5 \mid \text{superstar doesn't play}) = \binom{5}{4} \cdot 0.5^5$

$= 0.15625$

Law of total Probability

$P(\text{win } 4/5) = 0.15625 * 0.25 + 0.36015 * 0.75 = 0.309175$

Bayes Thm

$P(\text{superstar plays} \mid \text{wins } 4/5) = \dfrac{0.36015 * 0.75}{0.309175} = \boxed{0.8737}$